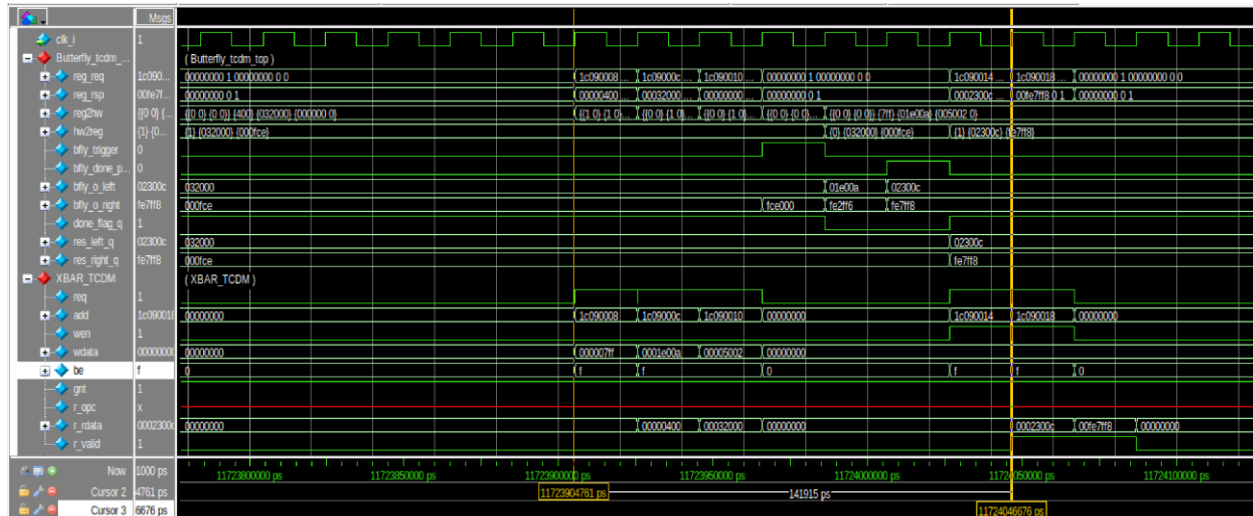




--- THE WRITE PHASE ---

- **Cycle 1 (The Twiddle Write):** The CPU executes `sw` (Store Word). You can see `req` goes high, `wen` goes low (0), and `add` becomes `...08`. The CPU pushes the twiddle index.
- **Cycle 2 (The Left Write):** The CPU executes the second `sw`. The `add` bus changes to `...0c`. The Left Operand is pushed.
- **Cycle 3 (The Right Write & Auto-Trigger):** The CPU executes the third `sw`. The `add` bus changes to `...10`. At the exact end of this clock cycle, look at your `bfly_trigger` signal—it pulses high!

--- THE NOP / COMPUTE PHASE ---

- **Cycle 4 (NOP 1):** The CPU executes the first `nop`. You can see `req` drops to 0. The bus is dead silent. Inside the silicon, your math core starts computing.
- **Cycle 5 (NOP 2):** The CPU executes the second `nop`. The bus remains silent. At the end of this cycle, look at `bfly_done_pulse`—it pulses high! The combinational math is finished.
- **Cycle 6 (NOP 3 & The Safety Latch):** The CPU executes the third `nop`. The bus remains silent. During this cycle, the `res_left_q` and `res_right_q` flip-flops finally catch up and latch the live math data safely into the registers.

--- THE READ PHASE ---

- **Cycle 7 (The Left Read):** The CPU executes `lw` (Load Word). The bus wakes up! `req` goes high, `wen` becomes 1 (Read), and `add` becomes `...14`.
- **Cycle 8 (The Right Read & Left Return):** The CPU executes the second `lw`, changing the address to `...18`. *At the exact same time*, because TCDM reads take 1 cycle, look at the bottom of the waveform: `r_valid` goes high and `r_rdata` returns the Left Result to the CPU.

- **Cycle 9 (The Final Return):** The CPU is done requesting, so `req` drops to 0. The TCDM bus returns the final Right Result (`r_valid` stays high, `r_rdata` updates).

(Add the 2 cycles of `perf` overhead, and you get exactly 11 cycles).